

Master Math for JEE Main & JEE Advanced

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JEE Mains 2020 Jan Chapter wise Question Bank

Ray Optics

Q1

Magnification of compound microscope is 375. Length of tube is 150mm. Given that focal length of objective lens is 5mm, then value of focal length of eyepiece is:

एक संयुक्त सूक्ष्मदर्शी की आवर्धन क्षमता 375 है। नली की लम्बाई 150 mm है। दिया गया है अभिदृश्यक लैन्स की फोकस दूरी $f_o = 5\text{mm}$ तब अभिनेत्र लेंस की फोकस दूरी f_e का मान होगा ?

- (1) 2mm (2) 22mm (3) 12mm (4) 33mm

7th Jan Morning

Sol

(2)

Case-I

If final image is at least distance of clear vision

यदि अन्तिम प्रतिबिम्ब निकटतम स्पष्ट दृष्टि पर हो तो

$$M.P. = \frac{L}{f_o} \left(1 + \frac{D}{f_e} \right); 375 = \frac{150}{5} \left[1 + \frac{25}{f_e} \right]$$

$$\frac{375}{30} = 1 + \frac{25}{f_e}$$

$$\frac{345}{30} = \frac{25}{f_e}$$

$$f_e = \frac{750}{345} = 2.17 \text{ cm}; f_e \approx 22 \text{ mm}$$

Case-II

If final image is at infinity यदि अन्तिम प्रतिबिम्ब अनन्त पर हो

$$M.P. = \frac{L}{f_o} \left(\frac{D}{f_e} \right) = 375$$

$$f_e = 22 \text{ mm}$$

Q2

Focal length of convex lens in air is 16 cm ($\mu_{\text{glass}} = 1.5$). Now the lens is submerged in liquid of refractive index 1.42. Find the ratio of focal length in medium to focal length in air has closest value

हवा में उत्तल लेंस की फोकल दूरी 16 cm है ($\mu_{\text{glass}} = 1.5$) अब लेंस को 1.42 अपवर्तनांक वाले द्रव में डुबोया जाता है।

माध्यम तथा हवा में लेंस की फोकल दूरी का अनुपात का निकटतम मान होगा

- (1) 9 (2) 17 (3) 1 (4) 5

Sol

(1)

$$\frac{1}{f_a} = \left(\frac{\mu_g}{\mu_a} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f_m} = \left(\frac{\mu_g}{\mu_m} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{f_a}{f_m} = \frac{\left(\frac{\mu_g}{\mu_m} - 1 \right)}{\left(\frac{\mu_g}{\mu_a} - 1 \right)} = \frac{\left(\frac{1.50}{1.42} - 1 \right)}{\left[\frac{1.50}{1} - 1 \right]} = \frac{0.08}{(1.92)(0.5)}$$

$$\frac{f_m}{f_a} = \frac{(1.42)(0.5)}{0.08} = 8.875 \approx 9$$

Q3

A telescope has magnifying power 5 and length of tube is 60cm then the focal length of eye piece is—

एक दूरदर्शी की आवर्धन क्षमता 5 तथा नलीका की लम्बाई 60cm हो तो अभिनेत्र लेंस की फोकस दूरी ज्ञात करो।

(1) 10cm

(2) 20cm

(3) 30cm

(4) 40cm

8th Jan Morning

Sol

(1)

$$m = \frac{f_o}{f_e}$$

$$5 = \frac{f_o}{f_e}$$

$$f_o = 5f_e$$

$$f_o + f_e = 60$$

$$6f_e = 60$$

$$f_e = 10$$

Q4

A plano-convex lens of radius of curvature 30 cm made of refractive index 1.5 is kept in air. Find its focal length (in cm).

8th Jan Morning

Sol

60 cm

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$R_1 = \infty$$

$$R_2 = -30 \text{ cm}$$

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{\infty} - \frac{1}{-30} \right)$$

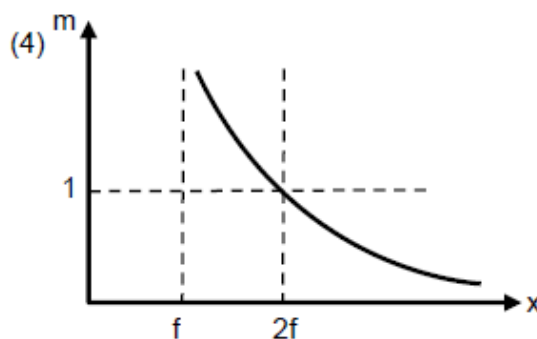
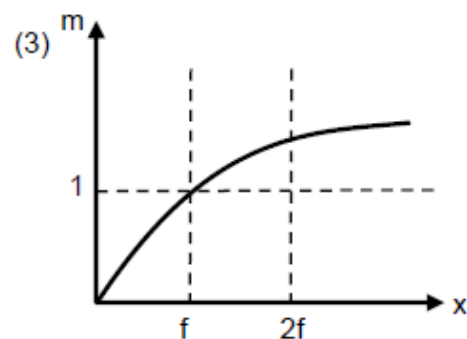
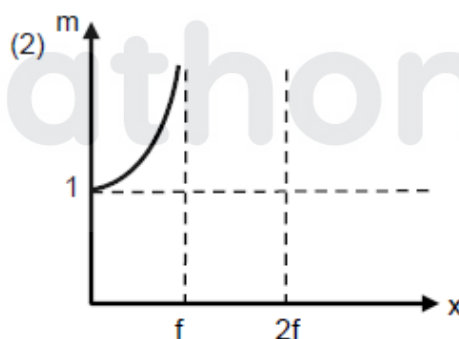
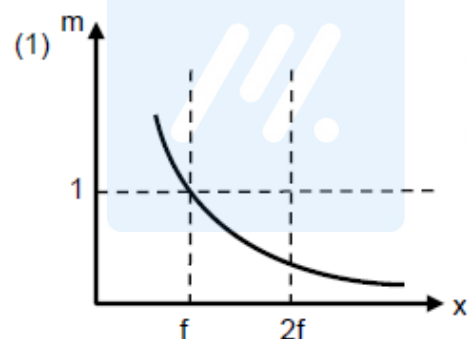
$$\frac{1}{f} = \frac{0.5}{30}$$

$$f = 60 \text{ cm}$$

Q5

An object is moving away from concave mirror of focal length f starting from focus. The distance of an object from pole of mirror is x . The correct graph of magnitude of magnification (m) versus distance x is:

एक वस्तु अवतल दर्पण के फोकस से मुख्य अक्ष के अनुदिश दर्पण से दूर जा रही है। वस्तु की दर्पण के ध्रुव से दूरी x है। आवर्धन (m) के परिमाण का दूरी x के साथ सही आरेख होगा :

8th Jan Evening

Sol

(4)

At focus, magnification is ∞ .

A telescope of aperture diameter 5m is used to observe the moon from the earth. Distance between the moon and earth is 4×10^5 km. Determine the minimum distance between two points on the moon's surface which can be resolved using this telescope. (Wave length of light is $5893 \text{ \AA}$).

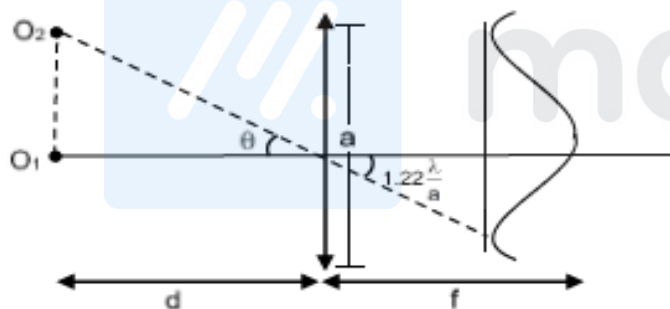
5m व्यास के द्वारक युक्त दूरदर्शी का प्रयोग पृथ्वी से चन्द्रमा को देखने के लिए करते हैं। पृथ्वी तथा चन्द्रमा के मध्य दूरी 4×10^5 km है। दूरदर्शी की सहायता से चन्द्रमा पर विभेदित किये जा सकने वाले दो बिन्दुओं के मध्य न्यूनतम दूरी लगभग कितनी होगी (प्रकाश की तरंगदैर्घ्य $5893 \text{ \AA}$ है)

- (1) 60 m (2) 20 m (3) 600 m (4) 200 m

9th Jan Morning

Sol

(1)



$$\theta = 1.22 \frac{\lambda}{a}$$

$$\text{distance दूरी} = O_1O_2 = d\theta$$

$$= 1.22 \frac{\lambda}{a} d$$

$$\text{distance दूरी} = O_1O_2 = \frac{1.22 \times 5893 \times 10^{-10} \times 4 \times 10^8}{5} \approx 57.5 \text{ m}$$

∴ answer from options विकल्पों के आधार पर उत्तर = 60m

(minimum distance न्यूनतम दूरी)

Q7

If a point source is placed at a depth h in a liquid of refractive index $\frac{4}{3}$. Find percentage of energy of light that escapes from liquid. (assuming 100% transmission of emerging light)

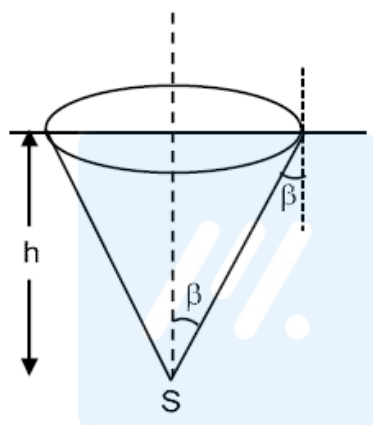
किसी द्रव में h गहराई पर एक बिन्दु स्रोत रखा है तथा द्रव का अपवर्तनांक $\frac{4}{3}$ हैं, तो द्रव से निकलने वाली ऊर्जा का प्रतिशत मान ज्ञात करो (बाहर निकलने वाले प्रकाश का प्रसिगमन 100% है)–

- (1) 15% (2) 17% (3) 21 % (4) 34%

9th Jan Evening

Sol

(2)



$$\sin \beta = \frac{3}{4}, \cos \beta = \frac{\sqrt{7}}{4}$$

$$\text{Solid angle वेस कोण } d\Omega = 2\pi R^2 (1 - \cos \beta)$$

$$\text{Percentage of light प्रकाश का प्रतिशत} = \frac{2\pi R^2 (1 - \cos \beta)}{4\pi R^2} \times 100$$

$$= \frac{1 - \cos \beta}{2} \times 100 = \left(\frac{4 - \sqrt{7}}{8} \right) \times 100 \approx 17\%$$

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